

Module 5 – C# IF, Switch, For, While loop Statements

The primary difference between a **for loop** and a **while loop** lies in how they are structured and used:

1. For Loop

- A **for loop** is typically used when the number of iterations is **known beforehand** (i.e., when you know exactly how many times the loop should run).
- The loop structure allows you to specify the initialization, condition, and increment/decrement all in one line.

```
for (initialization; condition; increment/decrement)
```

```
{
```

```
// Code to execute
```

```
}
```

```
for (int i = 1; i <= 5; i++)
```

```
{
```

```
    Console.WriteLine(i);
```

```
}
```

- Initialization: `int i = 1` – Starts the loop with `i` set to 1.
- Condition: `i <= 5` – The loop runs as long as `i` is less than or equal to 5.
- Increment/Decrement: `i++` – Increments `i` by 1 after each iteration.

While Loop

- A **while loop** is better suited for situations where the number of iterations is **unknown** and depends on a condition being met.
- The loop will keep running as long as the specified condition is `true`. It's often used when the loop should continue until a certain condition changes.

`while (condition)`

`{`

`// Code to execute`

`}`

```
int i = 1;

while (i <= 5)

{

    Console.WriteLine(i);

    i++;

}
```

- **Condition:** The loop runs as long as `i <= 5` is true.
- **Increment/Decrement:** You manually increment or change the variable inside the loop (`i++`).

Key Differences:

1. Use Case:

- **For Loop:** Best for when you know the number of iterations in advance (e.g., iterating over a list of 10 items).
- **While Loop:** Best for when the number of iterations is not known and depends on dynamic conditions (e.g., waiting for user input or monitoring a condition).

2. Initialization and Update:

- **For Loop:** Initialization, condition checking, and updating happen in one line.

- **While Loop:** Initialization is usually done separately, and updating must be handled inside the loop.

3. Readability:

- **For Loop:** More concise and readable when the number of iterations is predetermined.
- **While Loop:** Offers more flexibility for complex conditions where the end point is not easily known upfront.